

Simulação e Otimização

Optimization Mini-Project

Academic Year 2025/2026

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Objectives:

- To develop alternative methods (both metaheuristic and exact) for solving an optimization problem.
- To solve a given problem instance with the developed optimization methods.
- To analyze the solutions and running times obtained by the different methods.

Proposed optimization problem:

- Consider a Software Defined Network with a data plane defined by a graph $G = (N, A)$ with switches identified as $i = 1, 2, \dots, |N|$. Each link has an associated delay l_{ij} .
- Consider an integer parameter $n < |N|$ and two delay parameters W_{max} and C_{max} . The aim is to select n switches to connect one SDN controller to each of them, guaranteeing that the shortest path delay between any switch and its closest SDN controller is not higher than W_{max} and the shortest path delay between any pair of SDN controllers is not higher than C_{max} . The candidate switches to connect SDN controllers is a given subset of N .
- The objective is to minimize the average shortest path delay from any switch to its closest SDN controller.

Problem instance:

- Solve the optimization problem for $n = 8$ nodes, $W_{max} = 700$ and $C_{max} = 500$ in the graph of the figure presented in the next page with $|N| = 200$ nodes and $|A| = 250$ links (the data associated with the graph is in files `Nodes200.txt`, `Links200.txt` and `L200.txt`). The subset of switches that can be connected to SDN controllers is highlighted in blue in the figure and is defined in the provided file `Candidates.txt`.

Tasks:

- Implement both GRASP (Greedy Randomized Adaptive Search Procedure) and GA (Genetic Algorithm) algorithms to solve the optimization problem with a stopping criterion based on running time. Through testing with the given problem instance, select the best settings for each metaheuristic algorithm.
- Solve the optimization problem with both metaheuristic algorithms. Run 10 times each algorithm with the best settings and a runtime limit of 60 seconds on each run. Report the minimum, average and maximum objective values obtained among all 10 runs for each algorithm.
- Implement and solve the optimization problem with Integer Linear Programming. Run 'Ipsolve' with a timeout of 10 minutes. Register the solution found, its objective value and the running time of 'Ipsolve'.
- Compare the results and running times obtained by all methods and draw all relevant conclusions.

Deliveries and deadlines:

A report must be written by each group (maximum 2 students) describing:

- the name and student number of each student of the group, and the effort percentage of each student in the development of the mini-project,

- the MATLAB code, duly explained, that generates the LP file used to solve the optimization problem with 'lpsolve',
- the MATLAB codes, duly explained, of both metaheuristic algorithms,
- the identification and justification of the best settings used by each metaheuristic algorithm,
- the results obtained by all optimization methods,
- the comparative analysis of the results obtained by all optimization methods.

A demonstration of the mini-project must be conducted in the practical part of the last class (2nd of June) with the presence of both students in each group (this demonstration counts for evaluation). A compressed file with the report (in PDF format) and with all MATLAB implementations must be submitted in 'elearning' until the end of 7th of June.

Evaluation criteria:

- Correct implementation of GRASP and proposed best settings based on testing (20%)
- Correct implementation of GA and proposed best settings based on testing (20%)
- Correct implementation of the exact method (20%)
- Strategies that go beyond the standard implementations of GRASP and/or GA (10%).
- Demonstration in the last class (10%)
- Report description of results, comparison analysis and drawn conclusions (10%)
- Quality and clarity of the report (10%)

